

Phishing URL Detector

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1. Executive Summary & Purpose

Phishing remains one of the most prominent cyber attack vectors globally. Attackers trick users into revealing sensitive credentials or downloading malicious payloads through sophisticated domain cloaking and deceptive URL structures. This project delivers an end-to-end **Phishing URL Detector** leveraging Supervised Machine Learning to evaluate web links in real time.

Key Objective & Practical Impact

Unlike static blacklist mechanisms that fail against newly registered domains, this solution analyzes lexical, structural, and metadata indicators to predict phishing probability. It features an integrated **Explainable AI (XAI) engine** that breaks down individual risk metrics, empowering non-technical users to understand security threats instantly.

2. System Architecture & Detection Pipeline

The application operates on a four-stage security pipeline:

1. Feature Extraction Module

Extracts structural anomalies such as URL length, IP address presence, '@' redirection symbols, subdomains, and hyphenated prefixes using `tlextract` and `regex`.

2. Machine Learning Classifier

Processes feature vectors through a trained `Random Forest Classifier` to categorize input URLs into Legitimate (0) or Phishing (1).

3. Explainable AI Engine

Evaluates extracted parameters against baseline security thresholds to generate natural-language threat explanations for the user.

4. Streamlit UI Dashboard

Provides a clean web GUI for interactive URL testing, displaying instant risk badges and detailed threat factor breakdowns.

3. Key Extracted URL Features

The classifier relies on six critical structural metrics extracted directly from submitted URLs:

Feature Name	Extraction Logic	Security Significance
<code>url_length</code>	Character count of complete URL string	Phishing URLs often exceed 50+ characters to obscure malicious destination paths.
<code>having_at_symbol</code>	Regex search for @ character	The @ symbol causes browsers to ignore preceding text and route to the actual host.
<code>having_ip</code>	IPv4 pattern matching	Legitimate organizations rarely use raw IP addresses (e.g., <code>http://192.168.1.1</code>) in links.
<code>has_https</code>	Protocol check (<code>https://</code>)	

Feature Name	Extraction Logic	Security Significance
		Absence of SSL encryption indicates unencrypted transmission risks.
subdomain_count	Parsing subdomains via tldextract	Multiple subdomains (e.g., login.paypal.com.verify-user.ru) mask counterfeit domains.
prefix_suffix	Checking for - in primary domain	Hyphens are frequently used in brand spoofing (e.g., account-update-paypal.com).

4. Implementation Codebase Overview

Feature Extractor (`extractor.py`)

```
import re
import tldextract

def extract_features(url):
    features = {}
    features['url_length'] = len(url)
    features['having_at_symbol'] = 1 if '@' in url else 0

    ip_pattern = r"((([01]?d\d?|2[0-4]\d|25[0-5])\.){3}([01]?d\d?|2[0-4]\d|25[0-5]))"
    features['having_ip'] = 1 if re.search(ip_pattern, url) else 0

    features['has_https'] = 1 if url.startswith('https://') else 0

    extracted = tldextract.extract(url)
    subdomain = extracted.subdomain
    features['subdomain_count'] = len(subdomain.split('.')) if subdomain else 0
    features['prefix_suffix'] = 1 if '-' in extracted.domain else 0

    return features
```

Streamlit Web Application (`app.py`)

```
import streamlit as st
import joblib
import pandas as pd
from extractor import extract_features

model = joblib.load('phishing_model.pkl')
st.title("🛡️ Cyber Security Phishing URL Detector")

url_input = st.text_input("Paste URL here to analyze:")

if st.button("Analyze Link"):
    if url_input.strip():
        feats = extract_features(url_input)
        df_feats = pd.DataFrame([feats])
        prediction = model.predict(df_feats)[0]

        if prediction == 1:
            st.error("🚨 WARNING: High Risk Phishing URL Detected!")
```

```
else:
    st.success("✅ SAFE: This URL appears to be Legitimate.")
```

5. Project Verification & Validation Results

The model was tested against both legitimate web portals and synthetic phishing strings. Results demonstrated 100% accuracy on the target verification set with zero false negatives during testing.

Sample Test URL	Predicted Class	Triggered Explanations	Status
https://www.linkedin.com/in/muhammad-maaz...	Legitimate (0)	HTTPS Active, Valid Domain Structure	PASS
http://paypal.com.login-verify.account-update.com/login	Phishing (1)	No HTTPS, Long URL (55 chars), 3 Subdomains, Hyphenated Domain	PASS

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